

Guilherme Ribeiro Figueira

Junior C++ Software Engineer

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Experience

C++ Software Engineer

Critical Techworks | BMW Group

Lisbon, Portugal (Hybrid)

Jan 2026 – present

- Developed safety-critical **C++17** real-time software for an ECU on **unannounced BMW vehicles**.
- **Embedded programming using RTOS** on ARM processors. Strict **memory safety** and **MISRA** compliant programming
- Contributed to multiple embedded platform applications such as **check control messages**, car **maintenance monitoring**, **customer personalization** of vehicle features and management of vehicle **DTCs**.
- Devised and implemented a **data-driven architecture** for an internal vehicle state management component.
- Contributed to the **middleware layer of a central ECU** with communication over message queues, SOME/IP and CAN.
- Maintained high test coverage via **TDD** and thorough **documentation** with **Sphinx**. Followed **Agile** and **Scrum** methodologies.

Unity XR Developer - Curricular Internship

Glartek

Leiria, Portugal (Hybrid)

Mar 2023 – Aug 2023

- Built an **AR 3D asset inspection tool** for a **Microsoft HoloLens 2** industry 4.0 project.
- Collaborated with the web team to design a pipeline for dynamic loading of 3D assets from backoffice into AR.
- Used linear algebra, graph search algorithms and caching to reliably process any 3D model with **optimal performance**.

Projects

Papoila Game Engine 🔗 | Game Engine | (C++17, ImGui, GLFW)

WIP C++17 **OpenGL game engine** with assimp model loading, textured mesh rendering, cubemap skyboxes, Phong lighting, editor camera controls, and proximity dithering built in.

Easy Dialog Editor 🔗 | Tool | (C++17, ImGui, SDL)

Lightweight **visual node editor** for **non-linear dialogue trees** in games. Exports dialogue data to JSON for engine agnostic integration. The entire program is a portable 2.7 MB executable with no external dependencies.

The Natural's Descent 🔗 | Game | (Unity, C#)

Implemented procedural cave generation with **Wave Function Collapse**, AI combat mechanics, and a data-oriented grid puzzle system for easy design. Worked in a team of 2.

The Shape Of Silence 🔗 | Game | (UE5, C++, Blueprint, FMOD)

Narrative puzzle game with C++ systems for interaction, inventory, item inspection, and **data-driven** puzzles via **Unreal Data Assets**, with **Blueprint** delegates and functions for level design. Non-linear dialogue authored with my own **EasyDialogueEditor** and **FMOD** audio.

More projects, further details, videos and source code can be found in my [Portfolio](#) 🔗 and [Github](#) 🔗.

Skills

Programming Languages: C++, C#, Python, LUA, Java, TypeScript, OCaml

Graphics: OpenGL, Vulkan, ImGui, GLFW, SDL3, WebGL, OpenCV, QGIS

Engines & Tools: Unreal Engine 5, Unity, FMOD, Blender, Photoshop, Audacity

Build & Documentation: Bazel, CMAKE, Doxygen, Sphinx, Jinja

Workflow tools: RenderDoc, Docker, Linux Shell, VIM, Git, Jira

Databases: PostgreSQL, SQLite

Languages: Portuguese (native), English (proficient)

Education

Nova School of Science and Technology

B.Sc. + M.Sc. in Computer Science and Engineering

Lisbon, Portugal

Sept 2020 – Jan 2025

- Master Thesis (**Grade: 19/20**): Developed a Unity MR authoring tool that dynamically transforms scanned rooms into tangible virtual environments using Quest 3 scene understanding, procedural placement rules, and tracked-object substitution.
- Implemented performant stencil-buffer portals for MR transitions, stress-tested with ~200 simultaneous portals on Quest 3 hardware running at a stable framerate.

Publications

Towards Augmented XR Movement and Interaction Confidence across Substitutional Reality

Jan 2025

6th place (out of 25) on public vote for best [presentation](#) 🔗

Guilherme R. Figueira, Rui Nóbrega

[10.1109/ICGI68463.2025.11302724](https://doi.org/10.1109/ICGI68463.2025.11302724) 🔗 (International Conference on Graphics and Interaction (ICGI), 2025)